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United States Patent Application Publication	20250280815
Kind Code	A1
Publication Date	September 11, 2025
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CARBON DIOXIDE-LURED MOSQUITO TRAPPING DEVICE

Abstract

A carbon dioxide-lured mosquito trapping device includes a main body, as well as a net cover assembly, a fan, and a mosquito storage box that are sequentially arranged from top to bottom in a height direction of the main body. The net cover assembly is provided with an air inlet part and an air outlet part, and under the action of the fan, air enters the mosquito storage box from the air inlet part and is finally discharged from the air outlet part. A carbon dioxide mosquito luring part configured for generating carbon dioxide gas arranged on a side of the main body is further included. The carbon dioxide mosquito luring part includes a discharge hole formed in the net cover assembly for outputting carbon dioxide gas. Despite of a simple overall structure and a low cost, the device of a simple structure achieves mosquito luring and trapping functions.

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Appl. No.: **19/002774**

Filed: **December 27, 2024**

Foreign Application Priority Data

CN	202420454969.5	Mar. 08, 2024

Related U.S. Application Data

parent WO continuation PCT/CN2024/084722 20240329 PENDING child US 19002774

Publication Classification

Int. Cl.: **A01M1/02** (20060101); **A01M1/06** (20060101); **A01M1/10** (20060101)

U.S. Cl.:

CPC **A01M1/023** (20130101); **A01M1/06** (20130101); **A01M1/106** (20130101); A01M2200/012 (20130101)

Background/Summary

CROSS REFERENCE TO THE RELATED APPLICATIONS [0001] This application is the continuation application of International Application No. PCT/CN2024/084722, filed on Mar. 29, 2024, which is based upon and claims priority to Chinese Patent Application No. 202420454969.5, filed on Mar. 8, 2024, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD
[0002] The present disclosure relates to the field of mosquito trapping devices, and in particular to a carbon dioxide-lured mosquito trapping device.
BACKGROUND
[0003] Mosquito traps are also known as mosquito capturing devices, mosquito exterminators, ecological mosquito traps, ecological mosquito exterminators, and ecological mosquito extermination systems.
[0004] As illustrated in patent documents with application Nos. 201921978518.7, 201921721518.9 and the like, mosquito traps of the prior art, to achieve good effects of mosquito trapping and extermination, usually have a range of attractant features such as odors, scents, sounds, colors, light, temperatures, humidity, electric shock, and the like, which have complex structures and extremely high costs. The most common and widely used mosquito capturing and exterminating devices are light-lured mosquito capturing and exterminating devices, which require use of mosquito-lured lamps or mosquito-exterminating lamps, and are still relatively complex in structure.
SUMMARY
[0005] An objective of the present disclosure is to provide a carbon dioxide-lured mosquito trapping device, so as to solve at least one of the above problems.
[0006] In an aspect, the present disclosure provides a carbon dioxide-lured mosquito trapping device, and the device includes a main body, as well as a net cover assembly, a fan, and a mosquito storage box sequentially arranged from top to bottom in a height direction of the main body, where the net cover assembly is provided with an air inlet part and an air outlet part, and under the action of the fan, air enters the mosquito storage box from the air inlet part and is finally discharged from the air outlet part; [0007] a carbon dioxide mosquito luring part configured for generating carbon dioxide gas arranged on a side of the main body is further included; and [0008] the carbon dioxide mosquito luring part at least includes a discharge hole formed in the net cover assembly for outputting carbon dioxide gas.
[0009] Therefore, the present disclosure provides a mosquito trapping device of a brand new structure, under the action of the fan of the mosquito trapping device, air enters the mosquito storage box sequentially through the air inlet part and is then blown out from the air outlet part, the air duct structure formed can be used to effectively capture surrounding mosquitoes into the mosquito storage box, and the fan is an only electricity utilization component for mosquito trapping, without any other complex mosquito trapping or exterminating facilities; and the carbon

dioxide mosquito luring part is arranged to simulate human respiration, and only carbon dioxide gas is available for mosquito luring, and is diffused under the action of the air duct, with a mosquito luring effect achieved. Despite of the simple overall structure and the low cost, the device of a simple structure achieves mosquito luring and trapping functions.

[0010] In some embodiments, the net cover assembly is arranged at an upper end part of the main body and forms a first accommodating cavity together with the main body, and the net cover assembly includes a first net cover, a second net cover, and a third net cover, where the first net cover is mounted at the upper end part of the main body, the second net cover is mounted around a middle part of the first net cover in a sleeved manner and located at an outer periphery of the fan, the third net cover is mounted on top of the second net cover and located directly above the fan, and the discharge hole is formed in the first net cover;

[0011] the air inlet part includes a first air inlet part arranged on the third net cover and matching the fan, and a second air inlet part arranged in the mosquito storage box and located directly below the fan; [0012] the air outlet part includes a first air outlet part arranged on the first net cover and communicated with the first accommodating cavity, and a third air outlet part arranged in the mosquito storage box and communicated with the first accommodating cavity; and [0013] when the fan works, air enters the mosquito storage box sequentially through the first air inlet part and the second air inlet part, and then is blown out from the first air outlet part after sequentially passing through the third air outlet part and the first accommodating cavity, which forms a first air duct.

[0014] In some embodiments, the air outlet part further includes a second air outlet part arranged on the second net cover and communicated with the first accommodating cavity, when the fan works, air enters the mosquito storage box sequentially through the first air inlet part and the second air inlet part, and then is blown out from the second air outlet part after sequentially passing through the third air outlet part and the first accommodating cavity, which forms a second air duct.

[0015] In this way, when the fan works, air enters the mosquito storage box sequentially through the first air inlet part and the second air inlet part, and then is blown out from the first air outlet part after sequentially passing through the third air outlet part and the first accommodating cavity, which forms the first air duct, and air enters the mosquito storage box sequentially through the first air inlet part and the second air inlet part, and then is blown out from the second air outlet part after sequentially passing through the third air outlet part and the first accommodating cavity, which forms the second air duct, such that when the fan of the device works, an air duct structure with a single air inlet passage and dual air outlet passages is formed. The air duct structure can be used to effectively capture surrounding mosquitoes into the mosquito storage box, and the fan is an only electricity utilization component for mosquito trapping, without any other complex mosquito trapping or exterminating facilities. Despite of a simple overall structure and a low cost, the device, due to a special structural design, achieves a mosquito trapping function. More importantly, carbon dioxide gas used for luring mosquitoes, under the action of the air duct, is quickly diffused to an upper end of the device, thereby significantly enhancing a mosquito luring effect.

[0016] In some embodiments, the first net cover is provided with a downwardly recessed groove, an upper end part of the second net cover is located in the groove, and the first air outlet part and the discharge hole are both arranged on a side wall of the groove. In this way, the groove is formed in the first net cover, which increases an air outlet area and reduces an air output force, and the first air outlet part and the discharge hole are arranged on the side wall of the groove, such that an air outlet direction is perpendicular to an air inlet direction, and an air outlet surrounds an air inlet, which not only avoids interference with the air inlet but also promotes the diffusion of carbon dioxide gas, thereby enhancing the luring effect.

[0017] In some embodiments, the carbon dioxide mosquito luring part includes a carbon dioxide reservoir and a pressure reducing valve, the carbon dioxide reservoir is mounted in the main body, the pressure reducing valve is mounted in the main body and matches an air outlet end of the carbon dioxide reservoir, and the pressure reducing valve is communicated with the discharge hole

or the first accommodating cavity.
[0018] In some embodiments, the carbon dioxide-lured mosquito trapping device further includes a control part, and the carbon dioxide mosquito luring part further includes a flow regulation switch and a solenoid valve, where the flow regulation switch is connected to the pressure reducing valve, the solenoid valve is connected to the flow regulation switch, and the control part is mounted in the main body and is electrically connected to the solenoid valve and the fan. In this way, the flow regulation switch is configured for regulating output flow, and the solenoid valve is configured for opening or closing the air outlet.
[0019] In some embodiments, the carbon dioxide-lured mosquito trapping device further includes a carbon dioxide concentration detector, and the carbon dioxide concentration detector is mounted in the first accommodating cavity and is electrically connected to the control part. In this way, the carbon dioxide concentration detector can be used to monitor a concentration of carbon dioxide gas.
[0020] In some embodiments, the carbon dioxide-lured mosquito trapping device further includes an alarm, the alarm is mounted in the main body and is electrically connected to the control part, and the alarm is configured for sounding an alarm when the carbon dioxide concentration detector detects that the concentration of carbon dioxide gas exceeds a set threshold. In this way, the alarm is capable of sounding an alarm when the carbon dioxide concentration detector detects that the concentration of carbon dioxide gas exceeds the set threshold, to remind a user to manually reduce gas output or turn it off; and when the carbon dioxide concentration detector detects that the concentration of carbon dioxide gas remains unchanged in both open and closed states, it is generally believed that a reserve in the carbon dioxide reservoir is insufficient and nearly used up.
[0021] In some embodiments, the carbon dioxide-lured mosquito trapping device further includes a side cover, an accommodating groove is formed on a side surface of the main body, the carbon dioxide reservoir is mounted in the accommodating groove, and the side cover is detachably connected to the main body and is configured for covering the accommodating groove. In this way, this design facilitates disassembly, inspection or replacement of the carbon dioxide reservoir.
[0022] In some embodiments, the discharge hole is the first air outlet part.
[0023] In some embodiments, the carbon dioxide-lured mosquito trapping device further includes a support and a support cover, the support is mounted in the main body and is provided with a second accommodating space for accommodating the mosquito storage box, the mosquito storage box is detachably mounted in the second accommodating space, the support cover is mounted above the support, and the fan is mounted above the support cover. In this way, the mosquito storage box is detachably arranged, which facilitates timely cleaning of captured mosquitoes in the box, and allows for repeated use.
[0024] In some embodiments, the support cover is provided with an upwardly extending first protruding part, the first protruding part forms an accommodating groove for accommodating the fan, a second protruding part extending downward is arranged in the second net cover, the second protruding part is connected with the first protruding part, and the second protruding part is located on an inner side of the second air outlet part. In this way, the first protruding part and the second protruding part enable to separate the air inlet passage from the air outlet passage, and this structure is also a main structure of forming the dual air outlet passages.
[0025] In some embodiments, the carbon dioxide-lured mosquito trapping device further includes a padding plate, the padding plate is mounted on the support cover and located directly below the fan, the padding plate and the first protruding part are enclosed to form the accommodating groove, and the padding plate is provided with a first through hole communicated with the accommodating groove. In this way, the padding plate plays a role of supporting the fan without affecting air input.
[0026] In some embodiments, the mosquito storage box includes a box body, a mesh member, a fixing member, and a cover body, the mesh member is mounted on an inner wall of the box body through the fixing member, the cover body is mounted at an upper end part of the box body, the

mesh member, the cover body and the box body are enclosed to form a second accommodating cavity, the second air inlet part is arranged on the cover body, and the third air outlet part is arranged on the box body and/or the mesh member.
[0027] The second air inlet part includes a second through hole formed in the cover body, and the third air outlet part includes a third through hole formed at a bottom of the box body and/or a mesh hole formed in the mesh member.
[0028] The mosquito storage box further includes a shielding part for shielding the second through hole, and the shielding part includes a pair of shielding sheets that can be rotatably mounted on the cover body.
[0029] When the fan does not work, the shielding sheets and the cover body are fitted to completely shield the second through hole, such that the second through hole is closed; and [0030] when the fan works, the shielding sheets, under the action of wind, are pushed away from the cover body, such that the second through hole is opened.
[0031] Therefore, the mosquito storage box of the present disclosure has a simple structure, which not only ensures that mosquitoes are effectively trapped inside the box body, but also effectively guarantees normal operation of the air ducts.
[0032] The present disclosure has the following beneficial effects: [0033] 1. when the fan works, air enters the mosquito storage box sequentially through the first air inlet part and the second air inlet part, and then is blown out from the first air outlet part after sequentially passing through the third air outlet part and the first accommodating cavity, which forms the first air duct, and air enters the mosquito storage box sequentially through the first air inlet part and the second air inlet part, and then is blown out from the second air outlet part after sequentially passing through the third air outlet part and the first accommodating cavity, which forms the second air duct, such that when the fan of the device works, an air duct structure with a single air inlet passage and dual air outlet passages is formed; the air duct structure can be used to effectively capture surrounding mosquitoes into the mosquito storage box, and the fan is an only electricity utilization component for mosquito trapping, without any other complex mosquito trapping or exterminating facilities; despite of a simple overall structure and a low cost, the device, due to a special structural design, achieves a mosquito trapping function; [0034] 2. the groove is formed in the first net cover, which increases an air outlet area and reduces an air output force, and the first air outlet part is arranged on a wall of the groove, such that an air outlet direction is perpendicular to an air inlet direction, and an air outlet surrounds an air inlet, which not only avoids interference with the air inlet; and [0035] 3. carbon dioxide gas used for luring mosquitoes, under the action of the air duct, is quickly diffused to an upper end of the device, thereby achieving a good mosquito luring effect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0036] FIG. **1** is a schematic diagram of a three-dimensional structure of a carbon dioxide-lured mosquito trapping device according to an embodiment of the present disclosure.
[0037] FIG. **2** is a schematic diagram of a three-dimensional structure of the carbon dioxide-lured mosquito trapping device illustrated in FIG. **1** with a partial structure omitted.
[0038] FIG. **3** is a schematic diagram of an overhead structure of the carbon dioxide-lured mosquito trapping device illustrated in FIG. **2**.
[0039] FIG. **4** is a schematic diagram of a sectional structure of the carbon dioxide-lured mosquito trapping device illustrated in FIG. **3** in an A-A direction.
[0040] FIG. **5** is a schematic diagram of a three-dimensional structure of the carbon dioxide-lured mosquito trapping device illustrated in FIG. **2** with a partial structure omitted.
[0041] FIG. **6** is an enlarged view of a part B of the carbon dioxide-lured mosquito trapping device

illustrated in FIG. **5**.

[0042] FIG. **7** is a schematic diagram I of an exploded structure of the carbon dioxide-lured mosquito trapping device illustrated in FIG. **2**.

[0043] FIG. **8** is a schematic diagram II of an exploded structure of the carbon dioxide-lured mosquito trapping device illustrated in FIG. **2**.

[0044] FIG. **9** is a schematic diagram of a three-dimensional structure of a second net cover of the carbon dioxide-lured mosquito trapping device illustrated in FIG. **8**.

[0045] Reference numerals in FIGS. **1-9**: **1**. main body; **2**. mosquito storage box; **3**. fan; **4**. net cover assembly; **5**. support; **6**. support cover; **7**. padding plate; **8**. carbon dioxide mosquito luring part; **9**. control part; **10**. power supply part; **12**. side cover; **13**. base; **4***a*. air inlet part; **4***b*. air outlet part; **41**. first net cover; **42**. second net cover; **43**. third net cover; **50**. second accommodating space; **81**. discharge hole; **82**. carbon dioxide reservoir; **83**. pressure reducing valve; **84**. flow regulation switch; **85**. solenoid valve; **86**. box body; **87**. exhaust pipe; **91**. control panel; **92**. control circuit board; **100**. first accommodating cavity; **200**. second accommodating cavity; **101**. power plug; **21**. second air inlet part; **300**. accommodating groove; **411**. first air outlet part; **412**. groove; **421**. second air outlet part; **422**. second protruding part; and **431**. first air inlet part.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0046] The present disclosure will be further described below with reference to the accompanying drawings and specific embodiments.

Example 1

[0047] FIGS. **1-9** schematically illustrate a carbon dioxide-lured mosquito trapping device according to an embodiment of the present disclosure.

[0048] As illustrated in FIGS. **1-9**, the carbon dioxide-lured mosquito trapping device includes a main body **1**, as well as a net cover assembly **4**, a fan **3**, and a mosquito storage box **2** that are sequentially arranged from top to bottom in a height direction of the main body **1**. The net cover assembly **4** is provided with an air inlet part **4***a* and an air outlet part **4***b*, and under the action of the fan **3**, air enters the mosquito storage box **2** from the air inlet part **4***a* and is finally discharged from the air outlet part **4***b;* [0049] a carbon dioxide mosquito luring part **8** configured for generating carbon dioxide gas arranged on a side of the main body **1** is further included; and [0050] the carbon dioxide mosquito luring part **8** at least includes a discharge hole **81** formed in the net cover assembly **4** for outputting carbon dioxide gas.

[0051] The net cover assembly **4** is arranged at an upper end part of the main body **1** and forms a first accommodating cavity **100** together with the main body **1**, and the net cover assembly **4** includes a first net cover **41**, a second net cover **42**, and a third net cover **43**, where the first net cover **41** is mounted at the upper end part of the main body **1**, the second net cover **42** is mounted around a middle part of the first net cover **41** in a sleeved manner and located at an outer periphery of the fan **3**, the third net cover **43** is mounted on top of the second net cover **42** and located directly above the fan **3**, and the discharge hole **81** is formed in the first net cover **41**; [0052] the air inlet part **4***a* includes a first air inlet part **431** arranged on the third net cover **43** and matching the fan **3**, and a second air inlet part **21** arranged in the mosquito storage box **2** and located directly below the fan **3**; [0053] the air outlet part **4***b* includes a first air outlet part **411** arranged on the first net cover **41** and communicated with the first accommodating cavity **100**, and a third air outlet part arranged in the mosquito storage box **2** and communicated with the first accommodating cavity **100**; and [0054] when the fan **3** works, air enters the mosquito storage box **2** sequentially through the first air inlet part **431** and the second air inlet part, and then is blown out from the first air outlet part **411** after sequentially passing through the third air outlet part and the first accommodating cavity **100**, which forms a first air duct.

[0055] The air outlet part **4***b* further includes a second air outlet part **421** arranged on the second net cover **42** and communicated with the first accommodating cavity **100**, when the fan **3** works, air enters the mosquito storage box **2** sequentially through the first air inlet part **431** and the second air inlet part **21**, and then is blown out from the second air outlet part **421** after sequentially passing through the third air outlet part and the first accommodating cavity **100**, which forms a second air duct.

[0056] In this way, when the fan **3** works, air enters the mosquito storage box **2** sequentially through the first air inlet part **431** and the second air inlet part **21**, and then is blown out from the first air outlet part **411** after sequentially passing through the third air outlet part and the first accommodating cavity **100**, which forms the first air duct, and air enters the mosquito storage box **2** sequentially through the first air inlet part **431** and the second air inlet part **21**, and then is blown out from the second air outlet part **421** after sequentially passing through the third air outlet part and the first accommodating cavity **100**, which forms the second air duct, such that when the fan **3** of the device works, an air duct structure with a single air inlet passage and dual air outlet passages is formed. The air duct structure can be used to effectively capture surrounding mosquitoes into the mosquito storage box **2**, and the fan **3** is an only electricity utilization component for mosquito trapping, without any other complex mosquito trapping or exterminating facilities. Despite of a simple overall structure and a low cost, the device, due to a special structural design, achieves a mosquito trapping function. More importantly, carbon dioxide gas used for luring mosquitoes, under the action of the air duct, is quickly diffused to an upper end of the device, thereby significantly enhancing a mosquito luring effect.

[0057] The first net cover **41** is provided with a downwardly recessed groove **412**, an upper end part of the second net cover **42** is located in the groove **412**, and the first air outlet part **411** and the discharge hole **81** are both arranged on a side wall of the groove **412**. In this way, the groove **412** is formed in the first net cover **41**, which increases an air outlet area and reduces an air output force, and the first air outlet part **411** and the discharge hole **81** are arranged on the side wall of the groove **412**, such that an air outlet direction is perpendicular to an air inlet direction, and an air outlet surrounds an air inlet, which not only avoids interference with the air inlet but also promotes the diffusion of carbon dioxide gas, thereby enhancing the luring effect.

[0058] The carbon dioxide mosquito luring part **8** includes a carbon dioxide reservoir **82** and a pressure reducing valve **83**, the carbon dioxide reservoir **82** is mounted in the main body **1**, the pressure reducing valve **83** is mounted in the main body **1** and matches an air outlet end of the carbon dioxide reservoir **82**, and the pressure reducing valve **83** is communicated with the discharge hole **81** or the first accommodating cavity **100**.

[0059] The carbon dioxide-lured mosquito trapping device further includes a control part **9**, and the carbon dioxide mosquito luring part **8** further includes a flow regulation switch **84** and a solenoid valve **85**, where the flow regulation switch **84** is connected to the pressure reducing valve **83**, the solenoid valve **85** is connected to the flow regulation switch **84**, and the control part **9** is mounted in the main body **1** and is electrically connected to the solenoid valve **85** and the fan **3**. The flow regulation switch **84** is configured for regulating output flow, and the solenoid valve **85** is configured for opening or closing the air outlet. The flow regulation switch **84** can be connected to the discharge hole **81** through a pipeline.

[0060] The carbon dioxide-lured mosquito trapping device further includes a carbon dioxide concentration detector (not shown in the figure), and the carbon dioxide concentration detector is mounted in the first accommodating cavity **100** and is electrically connected to the control part **9**. In this way, the carbon dioxide concentration detector can be used to monitor a concentration of carbon dioxide gas, which facilitates timely regulation or replacement.

[0061] The carbon dioxide-lured mosquito trapping device further includes an alarm, the alarm is mounted in the main body **1** and is electrically connected to the control part **9**, and the alarm is configured for sounding an alarm when the carbon dioxide concentration detector detects that the concentration of carbon dioxide gas exceeds a set threshold.

[0062] The control part **9** in this embodiment includes a control panel **91** and a control circuit board **92** that matches the control panel **91**, and the control circuit board **92** is electrically connected to all

electricity utilization components.
[0063] The concentration of carbon dioxide gas can be displayed on the control panel **91**.
[0064] The alarm is capable of sounding an alarm when the carbon dioxide concentration detector detects that the concentration of carbon dioxide gas exceeds the set threshold, to remind a user to manually reduce gas output or turn it off; and [0065] when the carbon dioxide concentration detector detects that the concentration of carbon dioxide gas displayed on the control panel **91** remains unchanged in both open and closed states, it is generally believed that a reserve in the carbon dioxide reservoir **82** is insufficient and nearly used up.
[0066] The carbon dioxide-lured mosquito trapping device further includes a side cover **12**, an accommodating groove **300** is formed on a side surface of the main body **1**, the carbon dioxide reservoir **82** is mounted in the accommodating groove **300**, and the side cover **12** is detachably connected to the main body **1** and is configured for covering the accommodating groove **300**. In this way, this design facilitates disassembly, inspection or replacement of the carbon dioxide reservoir **82**.
[0067] As illustrated in FIGS. **2-9**, a mounting surface of the control part **9** is flush with an upper end face of the main body **1**, resulting in an aesthetically pleasing appearance. A base **13** is arranged at a bottom of the main body **1**, the base **13** matches the side cover **12**, and an accommodating cavity is formed on one side of the base **13**, and together with part of the main body **1**, forms the accommodating groove **300**.
[0068] As illustrated in FIGS. **5-6**, the carbon dioxide mosquito luring part **8** further includes a box body **86** and an exhaust pipe **87**, the box body **86** is mounted in the main body **1** and is located behind the discharge hole **81**, the discharge hole **81** is communicated with the box body **86**, and the box body **86** is communicated with the flow regulation switch **84** through the exhaust pipe **87**. The carbon dioxide concentration detector is arranged in the box body **87**, which facilitates detection of carbon dioxide gas and protects the carbon dioxide concentration detector.
[0069] The present disclosure provides a mosquito trapping device of a brand new structure, under the action of the fan **3** of the mosquito trapping device, air enters the mosquito storage box **2** sequentially through the air inlet part **4***a* and is then blown out from the air outlet part **4***b*, the air duct structure formed can be used to effectively capture surrounding mosquitoes into the mosquito storage box **2**, and the fan **3** is an only electricity utilization component for mosquito trapping, without any other complex mosquito trapping or exterminating facilities; and the carbon dioxide mosquito luring part **8** is arranged to simulate human respiration, and only carbon dioxide gas is available for mosquito luring, and is diffused under the action of the air duct, with a mosquito luring effect achieved. Despite of the simple overall structure and the low cost, the device of a simple structure achieves mosquito luring and trapping functions.
[0070] In this embodiment, the first air outlet part **411**, the second air outlet part **421**, the first air inlet part **431** and the like can be various shapes of holes through which air can pass.
[0071] The carbon dioxide-lured mosquito trapping device further includes a support **5** and a support cover **6**, the support **5** is mounted in the main body **1** and is provided with a second accommodating space **50** for accommodating the mosquito storage box **2**, the mosquito storage box **2** is detachably mounted in the second accommodating space **50**, the support cover **6** is mounted above the support **5**, and the fan **3** is mounted above the support cover **6**. In this way, the mosquito storage box **2** is detachably arranged, which facilitates timely cleaning of captured mosquitoes in the box, and allows for repeated use.
[0072] The mosquito storage box **2** of this example has a structure of the prior art, and a specific structure thereof will not be described in detail here.
[0073] The support cover **6** is provided with an upwardly extending first protruding part, the first protruding part forms an accommodating groove for accommodating the fan **3**, a second protruding part **422** extending downward is arranged in the second net cover **42**, the second protruding part **422** is connected with the first protruding part, and the second protruding part **422** is located on an

inner side of the second air outlet part **421**. In this way, the first protruding part and the second protruding part **422** enable to separate the air inlet passage from the air outlet passage, and this structure is also a main structure of forming the dual air outlet passages.
[0074] The carbon dioxide-lured mosquito trapping device further includes a padding plate **7**, the padding plate **7** is mounted on the support cover **6** and located directly below the fan **3**, the padding plate **7** and the first protruding part are enclosed to form the accommodating groove, and the padding plate **7** is provided with a first through hole communicated with the accommodating groove. In this way, the padding plate **7** plays a role of supporting the fan **3** without affecting air input.
[0075] In this embodiment, the carbon dioxide-lured mosquito trapping device further includes a power supply part **10**, and the power supply part **10** includes at least a power plug **101**.
[0076] The second air inlet part **21** can be a second through hole formed at a top of the mosquito storage box **2**, and the third air outlet part can be a third through hole formed at a bottom or periphery of the mosquito storage box **2**.
[0077] Preferably, a sum of air output from the first air outlet part **411** of the first net cover **41** and the second air outlet part **421** of the second net cover **42** is more than six times air input from the first air inlet part **431** on the third net cover **43**. In this way, it is effectively ensured that the air output force is less than an air input force, thereby facilitating mosquito trapping.
[0078] Preferably, in this embodiment, the main body **1**, the side cover **12**, the mosquito storage box **2**, the fan **3**, the net cover assembly **4**, the control panel **91** of the control part **9**, and the power plug **101** of the power supply part **10** all have appearances of dark colors such as black. The dark-colored appearances can enhance the mosquito luring effect.
[0079] The present disclosure provides the mosquito trapping device of a brand new structure, when the fan **3** of the device works, the air duct structure with a single air inlet passage and dual air outlet passages is formed, the air duct structure can be used to effectively capture surrounding mosquitoes into the mosquito storage box **2**, and the fan **3** is the only electricity utilization component for mosquito trapping, without any other complex mosquito trapping or exterminating facilities. Despite of the simple overall structure and the low cost, the device, due to the special structural design, achieves the mosquito trapping function.
Example 2
[0080] A carbon dioxide-lured mosquito trapping device of this example has a structure basically the same as that of the carbon dioxide-lured mosquito trapping device in Example 1, and the only difference therebetween lies in that the pipeline between the flow regulation switch **84** and the discharge hole **81** is omitted, carbon dioxide gas is directly discharged into the first accommodating cavity **100**, the original discharge hole is removed, and the first air outlet part **411** of this example is the discharge hole **81**. The carbon dioxide concentration detector is directly mounted in the first accommodating cavity **100** of the box.
[0081] In this way, carbon dioxide gas generated will directly enter the first accommodating cavity **100**, under the action of the fan **3**, then enters the groove **412** through the first air inlet, and under the action of the first air duct and the second air duct, is diffused to an upper end face of the device, thereby achieving circulation flow and a good mosquito luring effect.
[0082] The above are merely some embodiments of the present disclosure. For those of ordinary skill in the art, they may also make several transformations and improvements on the premise of not deviating from the inventive concept of the present disclosure, and these transformations and improvements shall fall within the scope of protection of the present disclosure.

Claims

1. A carbon dioxide-lured mosquito trapping device, comprising a main body, a net cover assembly, a fan, a mosquito storage box and a carbon dioxide mosquito luring part, wherein the net cover assembly, the fan and the mosquito storage box are sequentially arranged from top to bottom in a height direction of the main body, the net cover assembly is provided with an air inlet part and an air outlet part, and under an action of the fan, air enters the mosquito storage box from the air inlet part and is finally discharged from the air outlet part; the carbon dioxide mosquito luring part is configured for generating carbon dioxide gas and arranged on a side of the main body; the carbon dioxide mosquito luring part comprises a discharge hole formed in the net cover assembly for outputting the carbon dioxide gas; the net cover assembly is arranged at an upper end part of the main body and forms a first accommodating cavity together with the main body, and the net cover assembly comprises a first net cover, a second net cover, and a third net cover, wherein the first net cover is mounted at the upper end part of the main body, the second net cover is mounted around a middle part of the first net cover in a sleeved manner and located at an outer periphery of the fan, the third net cover is mounted on a top of the second net cover and located directly above the fan, and the discharge hole is formed in the first net cover; the air inlet part comprises a first air inlet part arranged on the third net cover and matching the fan, and a second air inlet part arranged in the mosquito storage box and located directly below the fan; the air outlet part comprises a first air outlet part arranged on the first net cover and communicated with the first accommodating cavity, and a third air outlet part arranged in the mosquito storage box and communicated with the first accommodating cavity; when the fan works, the air enters the mosquito storage box sequentially through the first air inlet part and the second air inlet part, and then is blown out from the first air outlet part after sequentially passing through the third air outlet part and the first accommodating cavity, wherein a first air duct is formed; and the carbon dioxide mosquito luring part comprises a carbon dioxide reservoir and a pressure reducing valve, wherein the carbon dioxide reservoir is mounted in the main body, the pressure reducing valve is mounted in the main body and matches an air outlet end of the carbon dioxide reservoir, and the pressure reducing valve is communicated with the discharge hole or the first accommodating cavity.

2. The carbon dioxide-lured mosquito trapping device according to claim 1, wherein the air outlet part further comprises a second air outlet part arranged on the second net cover and communicated with the first accommodating cavity, wherein when the fan works, the air enters the mosquito storage box sequentially through the first air inlet part and the second air inlet part, and then is blown out from the second air outlet part after sequentially passing through the third air outlet part and the first accommodating cavity, wherein a second air duct is formed.

3. The carbon dioxide-lured mosquito trapping device according to claim 2, wherein the first net cover is provided with a downwardly recessed groove, an upper end part of the second net cover is located in the downwardly recessed groove, and the first air outlet part and the discharge hole are arranged on a side wall of the downwardly recessed groove.

4. The carbon dioxide-lured mosquito trapping device according to claim 1, further comprising a control part, wherein the carbon dioxide mosquito luring part further comprises a flow regulation switch and a solenoid valve, wherein the flow regulation switch is connected to the pressure reducing valve, the solenoid valve is connected to the flow regulation switch, and the control part is mounted in the main body and is electrically connected to the solenoid valve and the fan.

5. The carbon dioxide-lured mosquito trapping device according to claim 4, further comprising a carbon dioxide concentration detector, wherein the carbon dioxide concentration detector is mounted in the first accommodating cavity and is electrically connected to the control part.

6. The carbon dioxide-lured mosquito trapping device according to claim 5, further comprising an alarm, wherein the alarm is mounted in the main body and is electrically connected to the control part, and the alarm is configured for sounding an alarm when the carbon dioxide concentration detector detects that a concentration of the carbon dioxide gas exceeds a set threshold.

7. The carbon dioxide-lured mosquito trapping device according to claim 1, further comprising a side cover, wherein an accommodating groove is formed on a side surface of the main body, the carbon dioxide reservoir is mounted in the accommodating groove, and the side cover is detachably

connected to the main body and is configured for covering the accommodating groove.
8. The carbon dioxide-lured mosquito trapping device according to claim 1, wherein the first air outlet part is the discharge hole.
9. The carbon dioxide-lured mosquito trapping device according to claim 2, wherein the first air outlet part is the discharge hole.
10. The carbon dioxide-lured mosquito trapping device according to claim 3, wherein the first air outlet part is the discharge hole.
11. The carbon dioxide-lured mosquito trapping device according to claim 4, wherein the first air outlet part is the discharge hole.
12. The carbon dioxide-lured mosquito trapping device according to claim 5, wherein the first air outlet part is the discharge hole.
13. The carbon dioxide-lured mosquito trapping device according to claim 6, wherein the first air outlet part is the discharge hole.
14. The carbon dioxide-lured mosquito trapping device according to claim 7, wherein the first air outlet part is the discharge hole.